Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: California Dairy Farms-Howard

Physical address of dairy:

2363 N Howard RD	Livingston	Merced	95334
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/01/1986

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X049-X090-X005-XXXX X049-X090-X006-XXXX X049-X090-X013-XXXX

B. OPERATORS

California Dairy Farms-Howard

Operator name: California Dairy Farms-Howard Telephone no.: (209) 285-0001 (Landline) (Cellular)

6071 Larson AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

California Dairyland LLC

Legal owner name: California Dairyland LLC Telephone no.: (209) 285-0001 (Landline) (Cellular)

6071 Larson AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	1,500	2,500	1,000	0
Number under roof	4,500	750	0	0	0	0
Maximum number	4,500	750	1,500	2,500	1,000	0
Average number	4,500	750	1,500	2,500	1,000	0
Avg live weight (lbs)	1,000	1,000	750	500		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 159,664.70 *tons per reporting period*

Total nitrogen from manure: 2,007,734.75 *lbs per reporting period*

After ammonia losses (30% loss applied): 1,405,414.33 *lbs per reporting period*

Total phosphorus from manure: 328,724.73 *lbs per reporting period*

Total potassium from manure: 832,226.52 *lbs per reporting period*

Total salt from manure: 2,291,287.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 49,695,000 *gallons*

Total nitrogen generated: 201,695.38 *lbs*

Total phosphorus generated: 23,637.36 *lbs*

Total potassium generated: 155,527.96 *lbs*

Total salt generated: 902,324.49 *lbs*

	49,695,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	49,695,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
6	31	31	2	process wastewater	X049-X009-X005-XXXX
7	26	26	2	process wastewater	X049-X009-X005-XXXX
Pivot 3	141	141	2	process wastewater	X049-X009-X005-XXXX
Pivot 4	136	136	2	process wastewater	X049-X009-X005-XXXX
Pivot 5	134	134	2	process wastewater	X049-X009-X004-XXXX X049-X009-X005-XXXX
Totals for areas that were used for application	468	468	10		
Totals for areas that were not used for application					
Land application area totals	468	468	10		

B. CROPS AND HARVESTS

6

Field name: 6

11/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	260.00 *ton*	As-is		66.8	5,800.00	800.00	3,500.00		8.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.39	97.29	13.42	58.71	448.86

6

06/30/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 31 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	300.00 *ton*	As-is		67.6	7,300.00	700.00	8,300.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.68	141.29	13.55	160.65	470.32

7

Field name: 7

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	240.00 *ton*	As-is		66.8	5,800.00	800.00	3,500.00		8.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.23	107.08	14.77	64.62	494.02

06/30/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 26 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	257.00 *ton*	As-is		67.6	7,300.00	700.00	8,300.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.88	144.32	13.84	164.08	480.39

Pivot 3

Field name: Pivot 3

11/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 141 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	2,065.00 *ton*	As-is		66.8	5,800.00	800.00	3,500.00		8.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.65	169.89	23.43	102.52	783.80

07/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 141 Plant date: 07/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	2,481.00 *ton*	As-is		70.0	6,400.00	400.00	13,800.00		16.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	17.60	225.23	14.08	485.64	1,752.54

Pivot 4

Field name: Pivot 4

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 136 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	1,873.00 *ton*	As-is		59.4	5,500.00	1,200.00	8,200.00		7.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.77	151.49	33.05	225.86	848.78

Pivot 4

07/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 136 Plant date: 07/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	3,783.00 *ton*	As-is		64.6	4,100.00	400.00	3,500.00		6.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.82	228.09	22.25	194.71	1,197.39

Pivot 5

Field name: Pivot 5

11/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 134 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	1,725.00 *ton*	As-is		59.4	5,500.00	1,200.00	8,200.00		7.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.87	141.60	30.90	211.12	793.38

07/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 134 Plant date: 07/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	1,904.00 *ton*	As-is		63.6	8,600.00	800.00	14,200.00		17.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	14.21	244.39	22.73	403.53	1,841.25

NUTRIENT BUDGET

A. LAND APPLICATIONS

6 - 11/29/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.04	7.92	69.70	596.29	378,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,185,280.00 *gal*
Application event totals		43.04	7.92	69.70	596.29	

12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.69	5.28	46.47	397.52	252,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,488,500.00 *gal*
Application event totals		28.69	5.28	46.47	397.52	

01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.84	10.33	60.06	231.55	672,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,836,890.00 *gal*
Application event totals		31.84	10.33	60.06	231.55	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.89	7.10	41.29	159.19	462,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,085,740.00 *gal*
Application event totals		21.89	7.10	41.29	159.19	

6 - 11/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.90	5.45	41.79	208.94	504,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,189,880.00 *gal*
Application event totals		28.90	5.45	41.79	208.94	

6 - 06/30/2023: Sudangrass, silage

Field name: 6

Crop: Sudangrass, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.94	7.73	59.20	295.99	714,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	5,474,700.00 *gal*
Application event totals		40.94	7.73	59.20	295.99	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.07	5.02	37.83	310.24	588,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	5,176,080.00 *gal*
Application event totals		30.07	5.02	37.83	310.24	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.85	4.48	33.78	277.00	525,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	5,474,700.00 *gal*
Application event totals		26.85	4.48	33.78	277.00	

6 - 06/30/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.11	7.53	56.75	465.36	882,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,943,820.00 *gal*
Application event totals		45.11	7.53	56.75	465.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.26	7.88	59.45	487.52	924,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,612,020.00 *gal*
Application event totals		47.26	7.88	59.45	487.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,981,600.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

7 - 11/30/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.51	5.24	46.17	394.98	210,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,820,300.00 *gal*
Application event totals		28.51	5.24	46.17	394.98	

7 - 11/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.77	7.87	69.26	592.46	315,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,355,780.00 *gal*
Application event totals		42.77	7.87	69.26	592.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.66	9.62	55.94	215.69	525,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,858,080.00 *gal*
Application event totals		29.66	9.62	55.94	215.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.10	8.47	49.23	189.80	462,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,687,580.00 *gal*
Application event totals		26.10	8.47	49.23	189.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.46	6.50	49.82	249.12	504,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,123,520.00 *gal*
Application event totals		34.46	6.50	49.82	249.12	

7 - 06/30/2023: Sudangrass, silage

Field name: 7

Crop: Sudangrass, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

7 - 06/30/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.63	7.86	60.20	301.02	609,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,777,920.00 *gal*
Application event totals		41.63	7.86	60.20	301.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.61	4.27	32.22	264.22	420,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,545,660.00 *gal*
Application event totals		25.61	4.27	32.22	264.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.58	5.77	43.49	356.69	567,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,114,320.00 *gal*
Application event totals		34.58	5.77	43.49	356.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.54	7.26	54.77	449.17	714,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,363,170.00 *gal*
Application event totals		43.54	7.26	54.77	449.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.23	8.55	64.44	528.43	840,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,185,280.00 *gal*
Application event totals		51.23	8.55	64.44	528.43	

7 - 06/30/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,649,800.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 3 - 11/18/2022: Oats, silage-soft dough

Field name: Pivot 3

Crop: Oats, silage-soft dough

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.09	11.60	102.16	873.99	2,520,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	15,344,640.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	11,229,840.00 *gal*
Application event totals		63.09	11.60	102.16	873.99	

01/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.25	6.25	36.31	140.00	1,848,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	10,229,760.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	7,486,560.00 *gal*
Application event totals		19.25	6.25	36.31	140.00	

02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.19	7.52	43.74	168.63	2,226,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	12,076,800.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,838,300.00 *gal*
Application event totals		23.19	7.52	43.74	168.63	

Pivot 3 - 11/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	14.87	4.83	28.06	108.18	1,428,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,808,960.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,446,760.00 *gal*
Application event totals		14.87	4.83	28.06	108.18	

04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.83	9.59	73.50	367.49	4,032,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	10,940,160.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,006,460.00 *gal*
Application event totals		50.83	9.59	73.50	367.49	

Pivot 3 - 07/05/2023: Corn, silage

Field name: Pivot 3

Crop: Corn, silage

Plant date: 07/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	265.87	6.86	14.92	6.01	846,000.00 *gal*
Application event totals		265.87	6.86	14.92	6.01	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	21,454,080.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	15,700,980.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 3 - 07/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	19,607,040.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	14,349,240.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	17,191,680.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	12,581,580.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	18,754,560.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	13,725,360.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 4 - 11/26/2022: Oats, silage-soft dough

Field name: Pivot 4

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.41	12.03	105.92	906.12	2,520,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	15,344,640.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	11,229,840.00 *gal*
Application event totals		65.41	12.03	105.92	906.12	

Pivot 4 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.96	6.47	37.65	145.14	1,848,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	10,229,760.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	7,486,560.00 *gal*
Application event totals		19.96	6.47	37.65	145.14	

02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.04	7.80	45.35	174.83	2,226,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	12,076,800.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,838,300.00 *gal*
Application event totals		24.04	7.80	45.35	174.83	

03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.42	5.00	29.09	112.16	1,428,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,808,960.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,446,760.00 *gal*
Application event totals		15.42	5.00	29.09	112.16	

04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.70	9.95	76.20	381.00	4,032,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	10,940,160.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,006,460.00 *gal*
Application event totals		52.70	9.95	76.20	381.00	

Pivot 4 - 07/06/2023: Corn, silage

Field name: Pivot 4

Crop: Corn, silage

Plant date: 07/06/2023

Pivot 4 - 07/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	265.87	6.86	14.92	6.01	816,000.00 *gal*
Application event totals		265.87	6.86	14.92	6.01	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	21,454,080.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	15,700,980.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	19,607,040.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	14,349,240.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	17,191,680.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	12,581,580.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	18,754,560.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	13,725,360.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 5 - 11/29/2022: Oats, silage-soft dough

Pivot 5 - 11/29/2022: Oats, silage-soft dough

Field name: Pivot 5

Crop: Oats, silage-soft dough

Plant date: 11/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.38	12.21	107.50	919.64	2,520,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	15,344,640.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	11,229,840.00 *gal*
Application event totals		66.38	12.21	107.50	919.64	

01/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.26	6.57	38.21	147.31	1,848,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	7,486,560.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	10,229,760.00 *gal*
Application event totals		20.26	6.57	38.21	147.31	

02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.40	7.92	46.02	177.44	2,226,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	12,076,800.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,838,300.00 *gal*
Application event totals		24.40	7.92	46.02	177.44	

03/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.65	5.08	29.52	113.83	1,428,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,808,960.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,446,760.00 *gal*
Application event totals		15.65	5.08	29.52	113.83	

Pivot 5 - 11/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.48	10.09	77.34	386.69	4,032,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	10,940,160.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,006,460.00 *gal*
Application event totals		53.48	10.09	77.34	386.69	

Pivot 5 - 07/07/2023: Corn, silage

Field name: Pivot 5

Crop: Corn, silage

Plant date: 07/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	265.87	6.86	14.92	6.01	804,000.00 *gal*
Application event totals		265.87	6.86	14.92	6.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	21,454,080.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	15,700,980.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	15,700,980.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	14,349,240.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Pivot 5 - 07/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	17,191,680.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	12,581,580.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

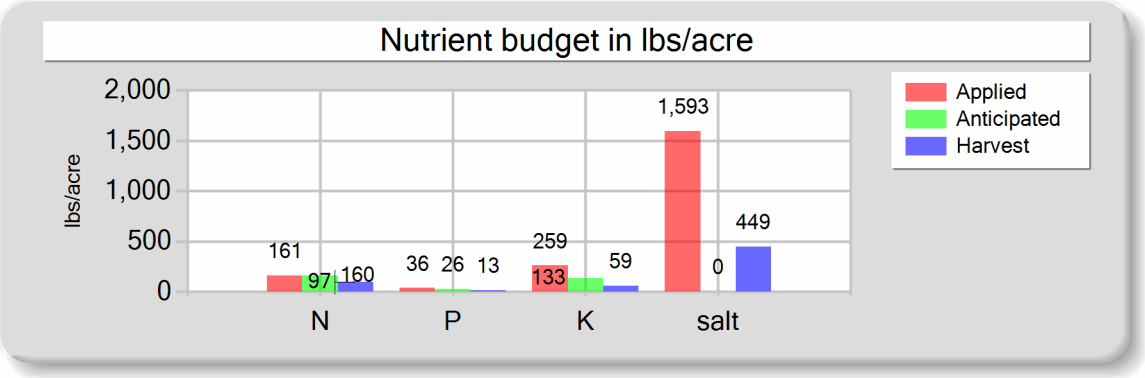
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	18,754,560.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	13,725,360.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

B. NUTRIENT BUDGET

6 - 11/29/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.36	36.08	259.31	1,593.49
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	161.36	36.08	259.31	1,593.49
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	97.29	13.42	58.71	448.86
Nutrient balance	64.07	22.66	200.60	1,144.62
Applied to removed ratio	1.66	2.69	4.42	3.55

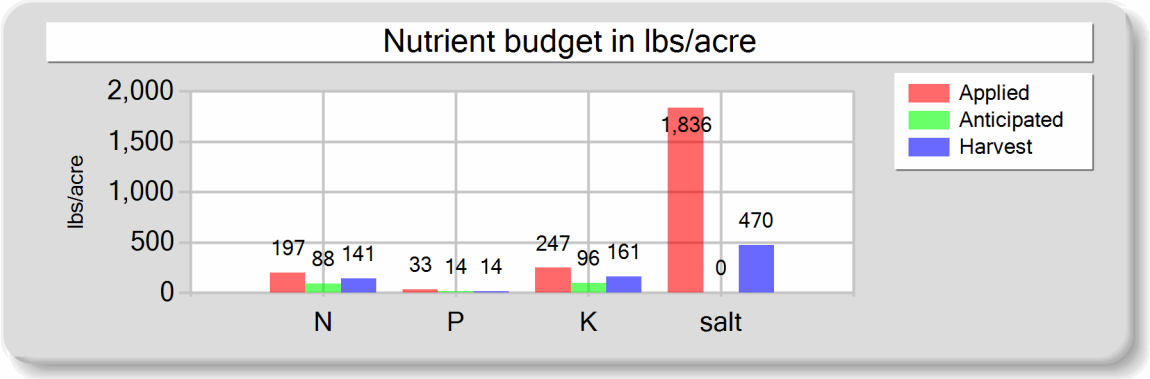
Fresh water applied	
13,786,290.00	*gallons*
507.70	*acre-inches*
16.38	*inches/acre*

Process wastewater applied	
2,268,000.00	*gallons*
83.52	*acre-inches*
2.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/30/2023: Sudangrass, silage

Field name: 6 Crop: Sudangrass, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.24	32.64	247.00	1,836.12
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.24	32.64	247.00	1,836.12
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	141.29	13.55	160.65	470.32
Nutrient balance	55.95	19.09	86.35	1,365.79
Applied to removed ratio	1.40	2.41	1.54	3.90

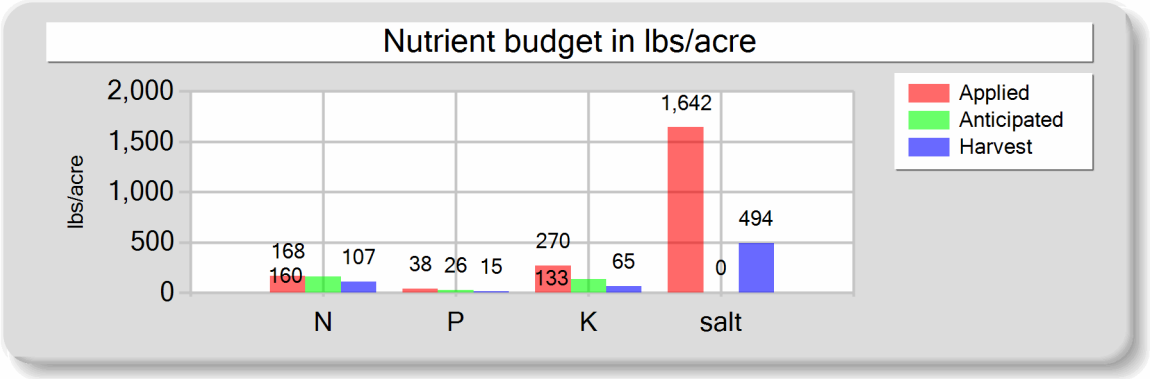
Fresh water applied	
29,662,920.00	*gallons*
1,092.38	*acre-inches*
35.24	*inches/acre*

Process wastewater applied	
3,633,000.00	*gallons*
133.79	*acre-inches*
4.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/30/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	161.49	37.70	270.42	1,642.05
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	168.49	37.70	270.42	1,642.05
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	107.08	14.77	64.62	494.02
Nutrient balance	61.41	22.93	205.81	1,148.03
Applied to removed ratio	1.57	2.55	4.19	3.32

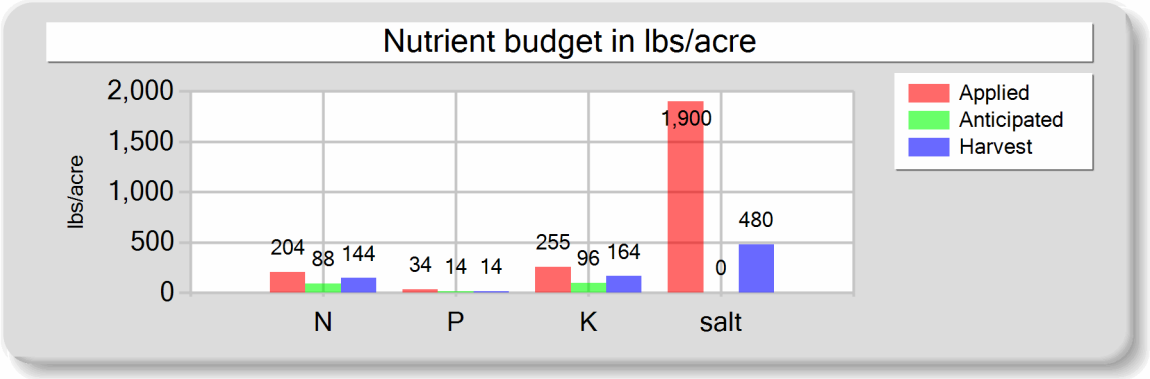
Fresh water applied	
11,845,260.00	*gallons*
436.22	*acre-inches*
16.78	*inches/acre*

Process wastewater applied	
2,016,000.00	*gallons*
74.24	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/30/2023: Sudangrass, silage

Field name: 7 Crop: Sudangrass, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.59	33.71	255.12	1,899.52
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.59	33.71	255.12	1,899.52
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	144.32	13.84	164.08	480.39
Nutrient balance	59.28	19.87	91.04	1,419.13
Applied to removed ratio	1.41	2.44	1.55	3.95

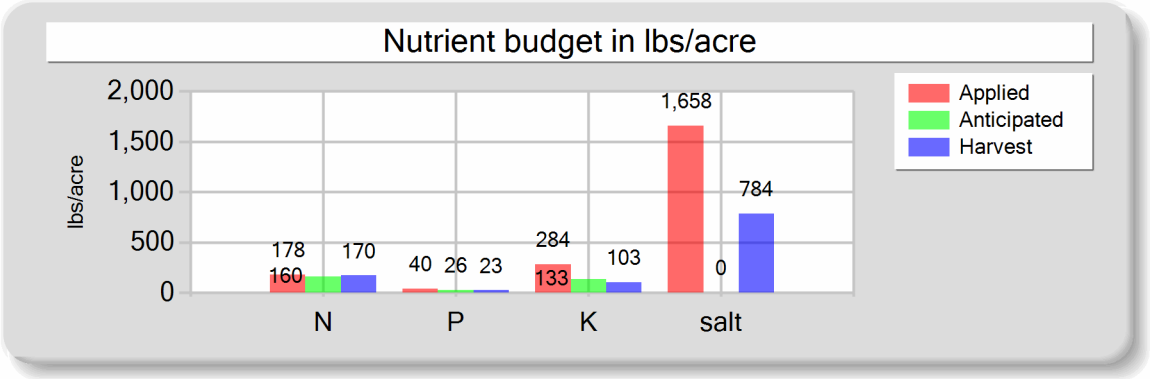
Fresh water applied	
24,636,150.00	*gallons*
907.27	*acre-inches*
34.89	*inches/acre*

Process wastewater applied	
3,150,000.00	*gallons*
116.00	*acre-inches*
4.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 3 - 11/18/2022: Oats, silage-soft dough

Field name: Pivot 3 Crop: Oats, silage-soft dough Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.23	39.79	283.77	1,658.29
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.23	39.79	283.77	1,658.29
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	169.89	23.43	102.52	783.80
Nutrient balance	8.34	16.36	181.25	874.49
Applied to removed ratio	1.05	1.70	2.77	2.12

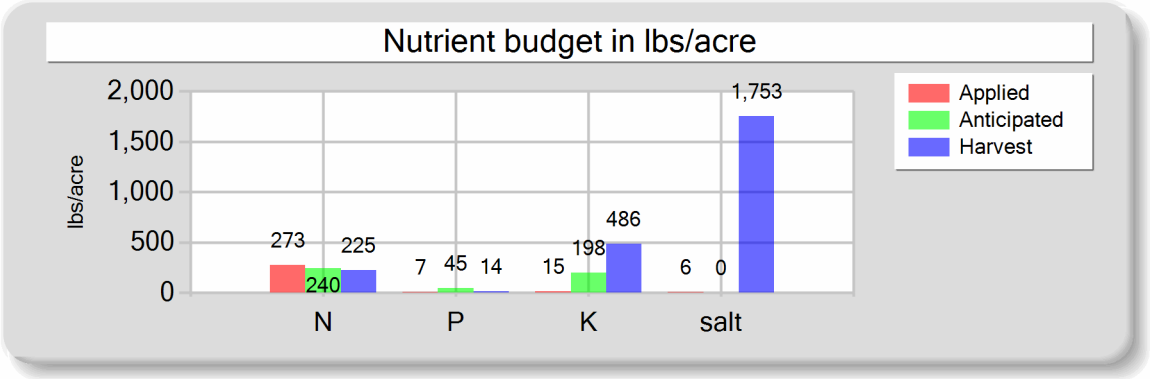
Fresh water applied	
99,408,240.00	*gallons*
3,660.87	*acre-inches*
25.96	*inches/acre*

Process wastewater applied	
12,054,000.00	*gallons*
443.91	*acre-inches*
3.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 3 - 07/05/2023: Corn, silage

Field name: Pivot 3 Crop: Corn, silage Plant date: 07/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.87	6.86	14.92	6.01
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.87	6.86	14.92	6.01
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.23	14.08	485.64	1,752.54
Nutrient balance	47.65	-7.22	-470.72	-1,746.53
Applied to removed ratio	1.21	0.49	0.03	0.00

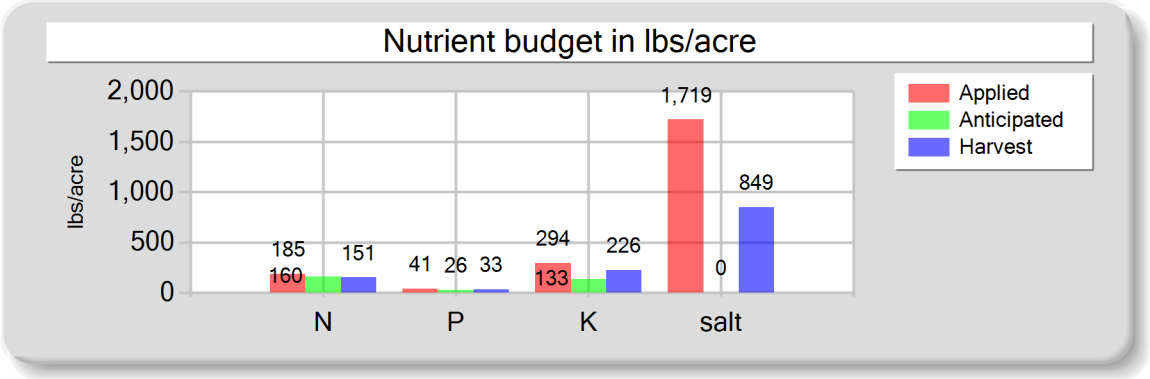
Fresh water applied	
133,364,520.00	*gallons*
4,911.36	*acre-inches*
34.83	*inches/acre*

Process wastewater applied	
846,000.00	*gallons*
31.16	*acre-inches*
0.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 4 - 11/26/2022: Oats, silage-soft dough

Field name: Pivot 4 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	177.52	41.25	294.21	1,719.26
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	184.52	41.25	294.21	1,719.26
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	151.49	33.05	225.86	848.78
Nutrient balance	33.03	8.20	68.34	870.47
Applied to removed ratio	1.22	1.25	1.30	2.03

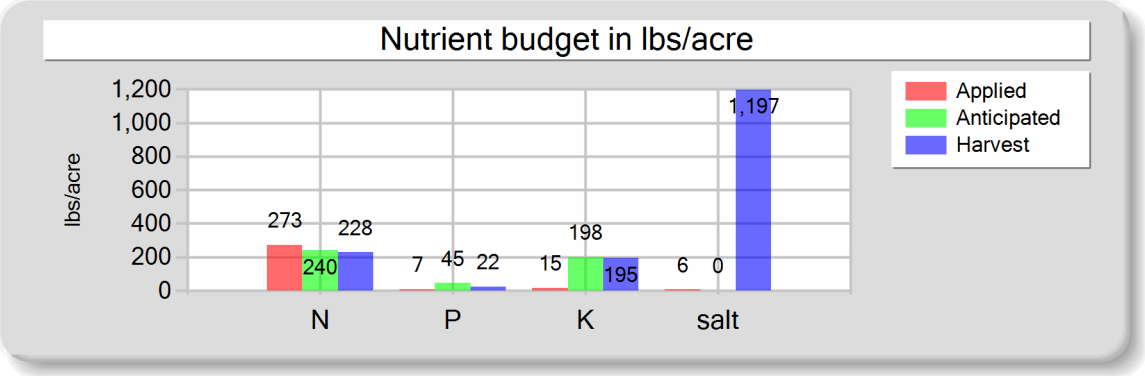
Fresh water applied	
99,408,240.00	*gallons*
3,660.87	*acre-inches*
26.92	*inches/acre*

Process wastewater applied	
12,054,000.00	*gallons*
443.91	*acre-inches*
3.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 4 - 07/06/2023: Corn, silage

Field name: Pivot 4 Crop: Corn, silage Plant date: 07/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.87	6.86	14.92	6.01
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.87	6.86	14.92	6.01
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	228.09	22.25	194.71	1,197.39
Nutrient balance	44.78	-15.39	-179.79	-1,191.38
Applied to removed ratio	1.20	0.31	0.08	0.01

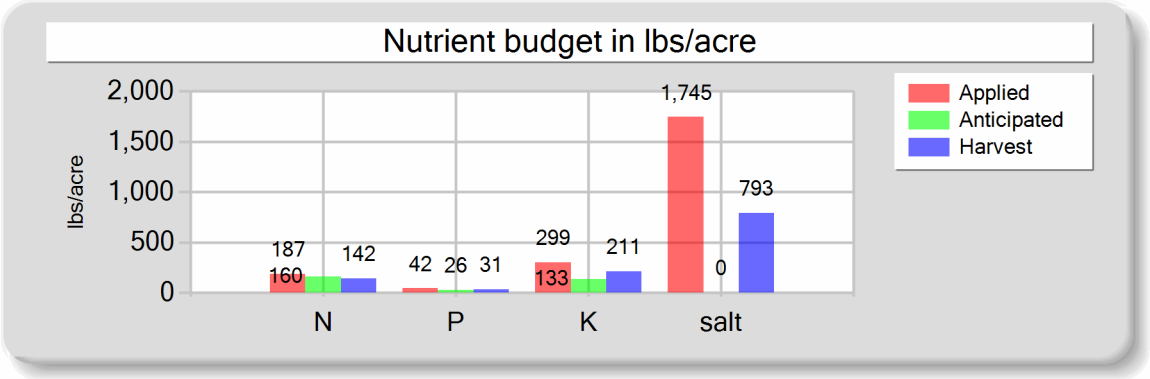
Fresh water applied	
133,364,520.00	*gallons*
4,911.36	*acre-inches*
36.11	*inches/acre*

Process wastewater applied	
816,000.00	*gallons*
30.05	*acre-inches*
0.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 5 - 11/29/2022: Oats, silage-soft dough

Field name: Pivot 5 Crop: Oats, silage-soft dough Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	180.17	41.87	298.60	1,744.92
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.17	41.87	298.60	1,744.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.60	30.90	211.12	793.38
Nutrient balance	45.57	10.97	87.48	951.54
Applied to removed ratio	1.32	1.36	1.41	2.20

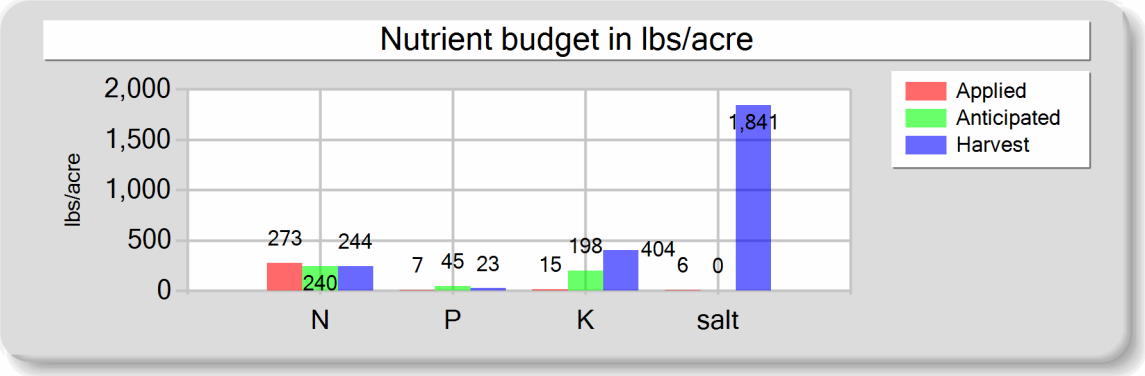
Fresh water applied	
99,408,240.00	*gallons*
3,660.87	*acre-inches*
27.32	*inches/acre*

Process wastewater applied	
12,054,000.00	*gallons*
443.91	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pivot 5 - 07/07/2023: Corn, silage

Field name: Pivot 5 Crop: Corn, silage Plant date: 07/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.87	6.86	14.92	6.01
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.87	6.86	14.92	6.01
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	244.39	22.73	403.53	1,841.25
Nutrient balance	28.48	-15.87	-388.61	-1,835.24
Applied to removed ratio	1.12	0.30	0.04	0.00

Fresh water applied	
129,458,460.00	*gallons*
4,767.51	*acre-inches*
35.58	*inches/acre*

Process wastewater applied	
804,000.00	*gallons*
29.61	*acre-inches*
0.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Sping Separated Solids

Sample and source description: Sping Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,600.00	4,100.00							17.50
DL	100.00	100.00	100.00							0.01

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,800.00	2,800.00	8,300.00							80.00
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 25.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	600.00	4,000.00	9,000.00	2,500.00	1,600.00	1,000.00	609.00		92.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	2,100.00	2,300.00	10,300.00	2,100.00	800.00	1,300.00	638.00		63.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	423.00	294.00	0.00	0.00	77.80	685.00								7,380.00	5,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	176.00	126.00	0.00	0.00	57.10	332.00								290.00	1,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 06/06/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5,310.00	546.00	0.00	0.00	137.00	298.00								1,920.00	120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	213.00	118.00	0.00	0.00	40.20	308.00								3,010.00	1,540
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	190.00	129.00	0.00	0.00	31.70	239.00	56.00	31.80	129.00	1,240.00	0.00	0.00	270.00	3,360.00	1,960
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 10/05/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	227.00	140.00	0.00	0.00	38.30	370.00								3,490.00	1,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Irrigation Well

IW # 7

Sample description: IW # 7

Sample date: 02/23/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,210.00	0
DL	0.50	0.50	0.50								1.00	10

IW #5

Sample description: IW #5

Sample date: 02/23/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								902.00	0
DL	0.50	0.50	0.50								1.00	10

IW #6

Sample description: IW #6

Sample date: 02/23/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								606.00	0
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

IW #9

Sample description: IW #9

Sample date: 02/23/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								616.00	0
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

6

Soil

Sample and source description: Soil

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			276.00				
DL			4.00				

7

Soil

Sample and source description: Soil

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			276.00				
DL			4.00				

Pivot 3

Pivot 3

Soil

Sample and source description: Soil

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			199.00				
DL			4.00				

Pivot 4

Soil

Sample and source description: Soil

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			118.00				
DL			4.00				

Pivot 5

Soil

Sample and source description: Soil

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			145.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

6 - 11/29/2022: Oats, silage-soft dough

6 - 11/29/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	1,300.00	12,300.00		11.50
DL	100.00	100.00	100.00		0.01

6 - 06/30/2023: Sudangrass, silage

Sudan Silage

Sample and source description: Sudan Silage

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,300.00	700.00	8,300.00		7.50
DL	100.00	100.00	100.00		0.01

7 - 11/30/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	1,300.00	12,300.00		11.50
DL	100.00	100.00	100.00		0.01

7 - 06/30/2023: Sudangrass, silage

Sudan Silage

Sample and source description: Sudan Silage

Sample date: 10/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,300.00	700.00	8,300.00		7.50
DL	100.00	100.00	100.00		0.01

Pivot 3 - 11/18/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,800.00	800.00	3,500.00		8.06
DL	100.00	100.00	100.00		0.01

Pivot 3 - 07/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	700.00	20,200.00		16.60
DL	100.00	100.00	100.00		0.01

Pivot 4 - 11/26/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,200.00	8,200.00		7.59
DL	100.00	100.00	100.00		0.01

Pivot 4 - 07/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,600.00	800.00	14,200.00		17.80
DL	100.00	100.00	100.00		0.01

Pivot 5 - 11/29/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	920.00	5,300.00		7.29
DL	100.00	100.00	100.00		0.01

Pivot 5 - 07/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	400.00	3,500.00		6.08
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

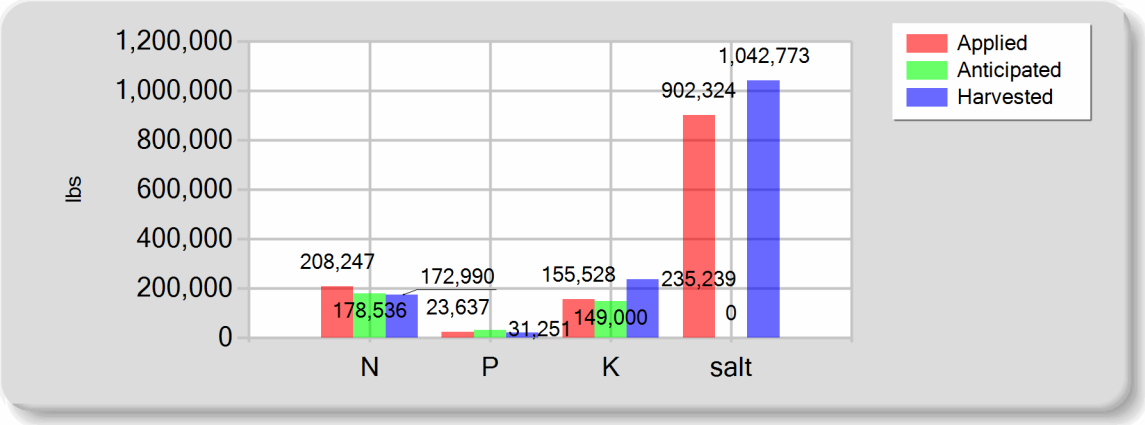
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

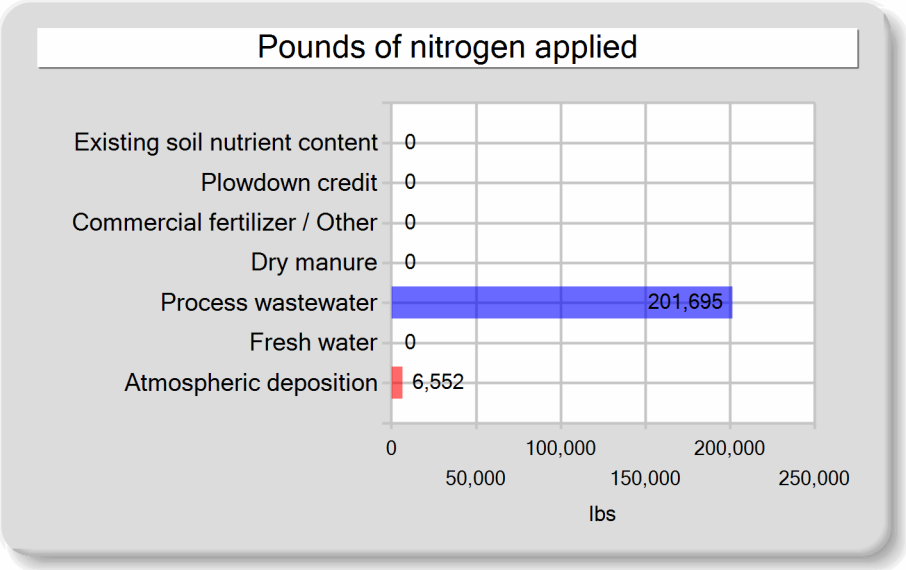
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

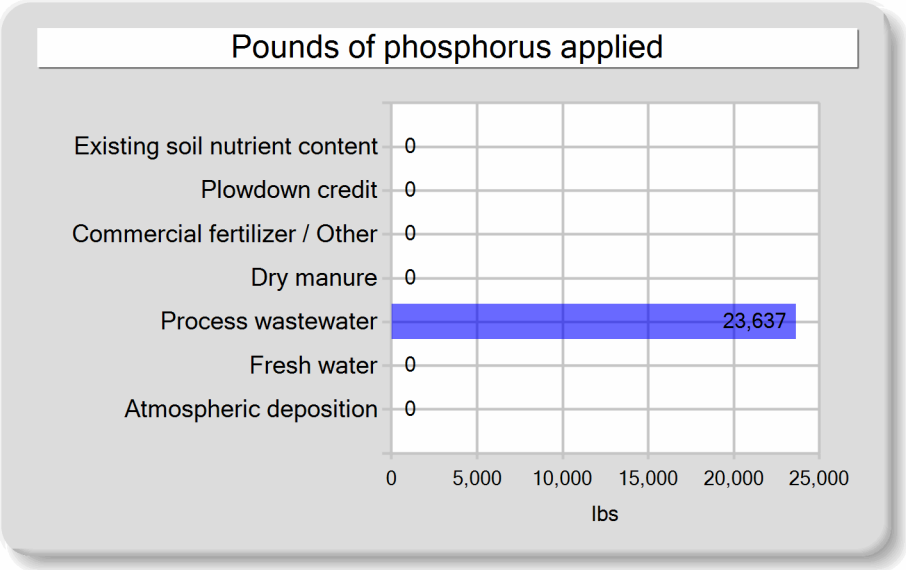
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201,695.38	23,637.36	155,527.96	902,324.49
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	6,552.00	0.00	0.00	0.00
Total nutrients applied	208,247.38	23,637.36	155,527.96	902,324.49
Anticipated crop nutrient removal	178,536.00	31,251.00	149,000.40	0.00
Actual crop nutrient removal	172,990.40	21,576.60	235,238.60	1,042,772.58
Nutrient balance	35,256.98	2,060.76	-79,710.64	-140,448.09
Applied to removed ratio	1.20	1.10	0.66	0.87

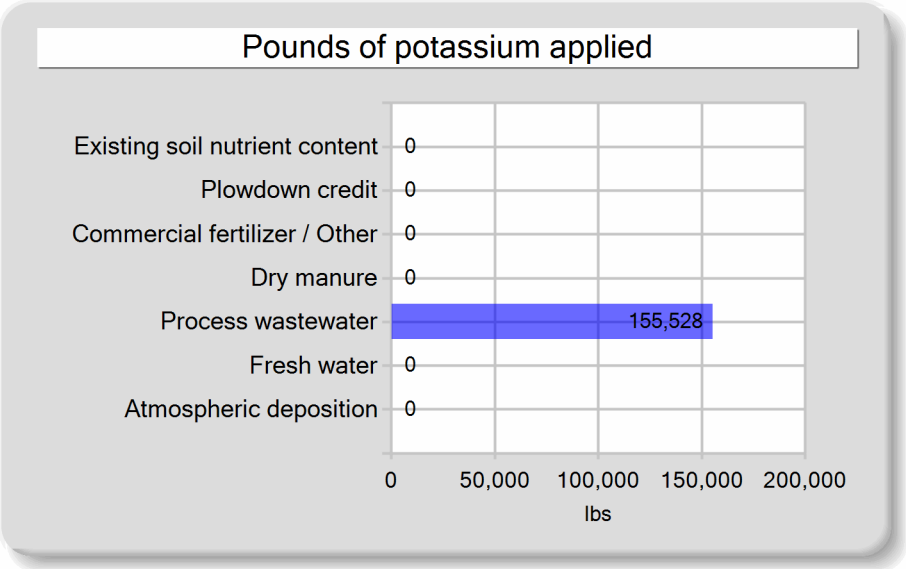
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

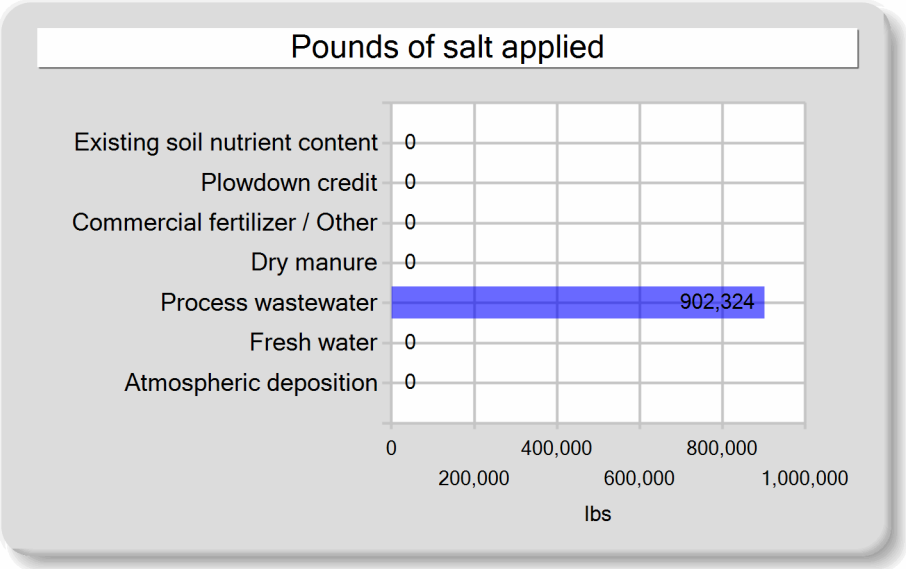


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 6 and 7 had lower than anticipated harvested tons (8.4-9.2 tons per acre harvested vs 16 tons per acre anticipated) resulted in these fields exceeding the 1.4 threshold. Application rates were in line with targeted removal and targeted harvested tons.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Larry A. Matos Larry A. Matos (Jun 21, 2024 14:33 PDT)	Larry A. Matos Larry A. Matos (Jun 21, 2024 14:33 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
California Dairyland LLC	California Dairy Farms-Howard
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 21, 2024	Jun 21, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
CDF Howard
P.O. Box 1090
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/23/24 9:05 AM
SUBMITTED BY: KR
ANALYZED DATE: 2/26/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Pivot 5 (IW #7)	T4054001A	Electrical Conductivity (EC)	EPA 120.1 and	1.21	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	7.82	mg/L
		Soluble Salts (SALT-SOL)	Calculation	772	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	110			0.764
Electrical Conductivity (EC)	96.1			0.248
Nitrate Nitrogen (NO3-N)	101	99.1 (T4054001A)	94.6 (T4054001A)	4.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Hilmar, CA 95324

Grower

Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/23/24 9:05 AM
SUBMITTED BY: KR
ANALYZED DATE: 2/26/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
585 Silage Well (IW #9)	T4054001B	Electrical Conductivity (EC)	EPA 120.1 and	0.616	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	394	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	110			0.764
Electrical Conductivity (EC)	96.1			0.248
Nitrate Nitrogen (NO3-N)	101	99.1 (T4054001A)	94.6 (T4054001A)	4.65

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/23/24 9:05 AM
SUBMITTED BY: KR
ANALYZED DATE: 2/26/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Feedlot Well (DW #4)	T4054001C	Electrical Conductivity (EC)	EPA 120.1 and	0.485	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	240	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	310	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	110			7.82 (T4054001A)	7.88 (null)	0.764
Electrical Conductivity (EC)	96.1			1.21 (T4054001A)	1.21 (Dup)	0.248
Nitrate Nitrogen (NO3-N)	101	99.1 (T4054001A)	94.6 (T4054001A)			4.65
Total Dissolved Solids (TDS)	94.0			240 (T4054001C)	255 (Dup)	6.06
Total Nitrogen (TN)	102			0.000 (T4054001C)	0.000 (null)	

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Customer
CDF Howard
P.O. Box 1090
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/23/24 9:05 AM
SUBMITTED BY: KR
ANALYZED DATE: 2/26/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
MB Well (DW #2/3)	T4054001D	Electrical Conductivity (EC)	EPA 120.1 and	0.499	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	190	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	< 1.50	mg/L
		Soluble Salts (SALT-SOL)	Calculation	319	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	110			7.82 (T4054001A)	7.88 (null)	0.764
Electrical Conductivity (EC)	96.1			1.21 (T4054001A)	1.21 (Dup)	0.248
Nitrate Nitrogen (NO3-N)	101	99.1 (T4054001A)	94.6 (T4054001A)			4.65
Total Dissolved Solids (TDS)	94.0			240 (T4054001C)	255 (Dup)	6.06
Total Nitrogen (TN)	102			0.000 (T4054001C)	0.000 (null)	

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Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/23/24 9:05 AM
SUBMITTED BY: KR
ANALYZED DATE: 2/26/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW #6 PV. 5 (IW #6)	T4054001E	Electrical Conductivity (EC)	EPA 120.1 and	0.606	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	388	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	110			0.764
Electrical Conductivity (EC)	96.1			0.248
Nitrate Nitrogen (NO3-N)	101	99.1 (T4054001A)	94.6 (T4054001A)	4.65

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Grower

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F & R Ag Services
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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 2/23/24 9:05 AM
SUBMITTED BY: KR
ANALYZED DATE: 2/26/24

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Weir Rd. (IW #5)	T4054001F	Electrical Conductivity (EC)	EPA 120.1 and	0.902	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.163	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	577	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	110			0.764
Electrical Conductivity (EC)	96.1			0.248
Nitrate Nitrogen (NO3-N)	101	99.1 (T4054001A)	94.6 (T4054001A)	4.65

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: CDF Howard
Mailing address: Livingston, CA
(not facility address)
Phone#:

Authorized Copy Released To: F&R Ag Services
Sampled By: Daniel N.
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/31	0500	Pivot 5 (IW#7)		12.0°c		X				
2		0510	58.5 silage well (IW#9)		12.0°c		X				
3		0610	Feed lot well (DW#4)		12.0°c						#7
4		0605	MB well (DW#2/3)		12.2°c						#7
5		0715	IW#6 Pr.5 (IW#6)		11.6°c		X				
6		0722	WEIRRD (IW#5)		11.6°c		X				
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	(KR)			2/23 905		
2nd		DENELE ANALYTICAL, INC.	2-23-24 9:05			T4054001

Required to be submitted to any agency? (YES / NO)

2023 Annual Report-CDF Howard

Final Audit Report 2024-06-21

Created:	2024-06-21
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAABacJeqwxbvShNh_7NzIPKR3WXE7S81J0

"2023 Annual Report-CDF Howard" History

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